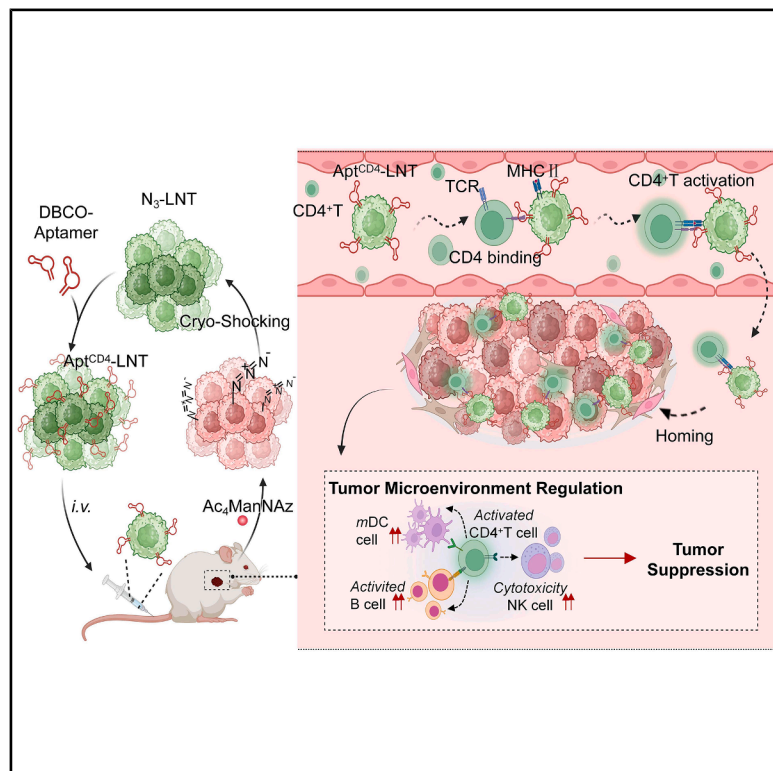


CD4 aptamer-engineered cell platforms enhance anti-tumor immunity in triple-negative breast cancer

Graphical abstract



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In brief

CD4⁺ T cells are crucial but underutilized anti-tumor warriors, hampered by poor tumor infiltration and immunosuppression. The Apt^{CD4}-LNTs platform overcomes this, selectively recruiting/activating CD4⁺ T cells in TNBC tumors. Activated CD4⁺ T cells recruit NK/B/DC cells, enhance MHC II immunity, and reshape the microenvironment, inhibiting TNBC tumors growth independently of CD8⁺ T cells. This strategy unlocks the full anti-tumor potential of CD4⁺ T cells, offering new avenues for TNBC immunotherapy.

Highlights

- CD4 aptamer captures and activates CD4⁺ T cells
- Apt^{CD4}-LNT drives *in vivo* infiltration and activation specific to CD4⁺ T cells
- Activated CD4⁺ T cells potently inhibits TNBC tumor growth by MHC II response
- CD4⁺ T cells remodel TNBC immunity by recruiting B, DC, and NK cells

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Article

CD4 aptamer-engineered cell platforms enhance anti-tumor immunity in triple-negative breast cancer

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THE BIGGER PICTURE Current immunotherapies' overreliance on CD8⁺ T cells fail patients with major histocompatibility complex class I (MHC I)-deficient cancers. Recently, CD4⁺ T cells offer a transformative solution: they help CD8⁺ T cells to kill MHC I⁺ tumors, directly kill MHC II⁺ tumors, and mobilize antigen-presenting cells/natural killer (NK) cells against MHC-deficient tumors. The therapeutic potential of CD4⁺ T cells has been confirmed in various types of cancer, including MHC-deficient melanomas, glioblastoma, and breast cancers. However, optimizing CD4⁺ T cell infiltration and activation within the tumor microenvironment remains a critical challenge to fully realizing their therapeutic potential. To overcome this, we developed a CD4 aptamer-engineered liquid nitrogen-treated cancer cell system (Apt^{CD4}-LNTs) to selectively activate and recruit CD4⁺ T cells for improved anti-tumor immunity in triple-negative breast cancer (TNBC). We revealed that CD4⁺ T cells reshape the tumor microenvironment and enhance MHC II anti-tumor immune response to inhibiting TNBC tumors. This strategy unlocks the full anti-tumor potential of CD4⁺ T cells, offering new avenues for TNBC immunotherapy and CD4⁺ T cell therapy.

SUMMARY

While CD8⁺ T cells are primary anti-tumor effectors, CD4⁺ T cells play critical helper and direct killer roles, yet their clinical use is hindered by poor tumor infiltration and the immunosuppressive tumor microenvironment (TME). To overcome this, we developed cluster of differentiation 4 (CD4) aptamer-engineered liquid nitrogen-treated cancer cells (Apt^{CD4}-LNTs) designed to selectively activate and recruit CD4⁺ T cells for improved anti-tumor immunity in triple-negative breast cancer (TNBC). This system leverages the CD4 aptamer's specificity for CD4⁺ T cells and the homing capacity of LNTs to drive targeted infiltration and activation of CD4⁺ T cells within the TNBC TME. Activated CD4⁺ T cells, in turn, enhance the infiltration and activity of natural killer (NK) cells, B cells, and dendritic cells, reshaping the immune landscape and potentiating anti-tumor responses. Our findings highlight the pivotal role of CD4⁺ T cells in TNBC immunotherapy and present a promising strategy for advancing CD4-targeted treatments.

INTRODUCTION

Immunotherapy has transformed cancer treatment in recent years, offering durable and long-lasting responses by harnessing

anti-tumor immune mechanisms.^{1,2} While CD8⁺ T cells are traditionally viewed as key effectors due to their ability to directly eliminate cancer cells,^{3,4} they often become exhausted in the immunosuppressive tumor microenvironment, diminishing their



cytotoxicity effectiveness.^{5,6} Additionally, some tumors evade CD8⁺ T cell recognition by downregulating major histocompatibility complex class I (MHC I), further limiting the efficacy of immunotherapy in certain patient populations.^{7,8}

To address these limitations, there has been increasing interest in engaging CD4⁺ T cells as an alternative or complementary approach.^{9–11} CD4⁺ T cells have diverse anti-tumor functions,¹⁰ including supporting other immune effectors such as CD8⁺ T cells and natural killer (NK) cells.^{12–14} Certain CD4⁺ T cell subsets can also directly kill MHC II-expressing tumor cells that are not recognized by CD8⁺ T cells.^{11,15,16} Additionally, CD4⁺ T cells can mobilize antigen-presenting cells, such as dendritic cells (DCs) and B cells, enabling them to combat MHC-deficient tumors.⁹

The therapeutic potential of CD4⁺ T cells has been demonstrated across various cancer types. For example, CD4⁺ T cell-mediated chimeric antigen receptor T cell therapy (CAR-T)¹⁷ and adoptive cell transfer⁹ have shown superior efficacy compared with CD8⁺ T cells in glioblastoma and MHC-deficient melanoma models, respectively. Similarly, in human epidermal growth factor receptor 2 positive (HER2⁺) breast cancer, CD4⁺ T cells were more effective in tumor suppression than CD8⁺ T cells.¹⁸ In addition, in triple-negative breast cancer (TNBC), CD4⁺ T cells alone were sufficient to initiate an immune response to suppress the tumor growth.¹⁹ However, optimizing CD4⁺ T cell infiltration and activation within the tumor microenvironment remains a critical challenge to fully realizing their therapeutic potential.^{9,20,21}

Aptamers, single-stranded DNA or RNA molecules, have emerged as powerful tools due to their high specificity, ease of production, and rapid tissue penetration.^{22,23} In addition to their ability to target specific molecules, aptamers can modulate immune functions, such as targeting immune checkpoints like programmed cell death ligand 1 (PD-L1)²⁴ and activating T cells through receptors like cluster of differentiation 3 (CD3) and CD28.^{25,26} While aptamers have been explored for improving cell infiltration and activation,²⁷ the selective use of aptamers to enhance CD4⁺ T cell infiltration has not been reported.

In this study, we present a novel strategy for selectively mobilizing CD4⁺ T cells using a CD4 aptamer-engineered liquid nitrogen-treated cancer cells system (Apt^{CD4}-LNTs). By leveraging the CD4 aptamer's specific targeting of CD4⁺ T cells and the tumor-homing capacity of cryo-shocked cancer cells,^{28,29} this approach facilitates selective CD4⁺ T cell infiltration into the tumor microenvironment, followed by their activation via CD4 aptamer. This strategy represents a potential avenue for enhancing the immunotherapy response of CD4⁺ T cells.

RESULTS

Design and preparation of Apt^{CD4}-LNTs

We selected aptamer #1-62, a previously reported CD4-specific aptamer,³⁰ as the CD4-targeting aptamer, hereafter referred to as Apt^{CD4} (Table S1; Figure S1A). Flow cytometry analysis confirmed that the CD4 aptamer (Apt^{CD4}) binds to CD4⁺ T cells in both mouse and human samples (Jurkat cells³¹), with significantly higher binding compared with the scrambled sequence control (Apt^{control}) and a non-specific aptamer library control (Apt^{Lib}) (Table S1; Figures 1A–1C and S1B–S1D). Moreover,

Apt^{CD4} showed no appreciable binding to CD4[−] Ramos cells, further confirming its specific recognition of CD4⁺ cells (Figures S1E and S1F). Direct CD4 purified protein binding was further validated by a significantly stronger binding signal from Apt^{CD4} compared with controls (Figure S1G). In degradation assays, polyacrylamide gel electrophoresis (PAGE) analysis revealed over 58% of unconjugated Apt^{CD4} remained intact over 24 h (Figures S1H and S1I), supporting its biochemical stability. Together, these results validated that Apt^{CD4} possesses both target specificity and stability to enable downstream applications.

Next, we used click chemistry to conjugate the aptamer to liquid nitrogen-treated cells (LNTs). Given the significance of CD4⁺ T cell in TNBC immunotherapy (Figure S2), the murine TNBC cell line 4T1 was selected for LNTs preparation. These cells were metabolically labeled with azide-modified glycans and cryo-inactivated through rapid freezing in liquid nitrogen (Figure 1D). To preserve cell structure and ensure the non-proliferative property of LNTs, we optimized cryo-shocking conditions of cryopreservation medium and freezing protocols. Key optimization criteria included cell viability (Figure S3A), proliferation rate, and structural integrity (Figures S3B and S3C). Under the optimal condition—using fetal bovine serum (FBS) as the cryopreservation medium and liquid nitrogen fast freezing (termed the M3-F condition)—LNTs were successfully prepared without observable cell proliferation or viability (Figures 1E and 1F), while maintaining their structural integrity (Figure 1G). Morphometric analysis by fluorescence imaging demonstrated that there was no significant difference in cell size between live cells and LNTs (Figures 1H and 1I).

Subsequently, a dibenzocyclooctyne (DBCO)-modified CD4 aptamer was then coupled to azide sugar engineered LNTs, generating Apt^{CD4}-LNTs (Figure 1J). Quantification analysis revealed that each LNT carried approximately $0.52 \times 10^{-6} \pm 0.03$ nmol of aptamers, corresponding to around 3.11×10^8 aptamer molecules per LNT (Figures S4A and S4B). The conjugated Apt^{CD4}-LNTs maintained binding to CD4 protein (Figure S4C) and the cellular-scale architecture for at least 72 h (Figure 1K).

To evaluate stability, we conducted both *in vitro* and *in vivo* analyses. Serum incubation revealed that 59.0% of free Apt^{CD4} and 87.7% of Apt^{CD4}-LNT conjugates remained intact after 24 h (Figures S4D and S4E). *In vivo*, 84.7% of Apt^{CD4}-LNTs was still detectable after 24 h (Figures S4F and S4G), indicating enhanced circulatory stability. Notably, the conjugated form demonstrated markedly improved stability over the free aptamer, likely due to structural protection afforded by conjugation, which shields aptamers from nuclease degradation and extends circulation time.³²

To further evaluate the safety of Apt^{CD4}-LNTs, *in vivo* tumorigenicity assays were performed. Unlike live cells, Apt^{CD4}-LNTs did not induce tumor growth (Figure 1L), demonstrating their safety profile *in vivo*. Collectively, these results confirmed the successful preparation of an Apt^{CD4}-LNTs system, with confirmed binding specificity, structural stability, and *in vivo* safety.

Apt^{CD4}-LNTs promoted CD4⁺ T cells infiltration

Next, we sought to determine whether Apt^{CD4}-LNTs selectively capture CD4⁺ T cells and further achieve the infiltration of CD4⁺ T cells through its homing capacity (Figure 2A). Tumor self-homing of Apt^{CD4}-LNTs is the key to achieving CD4⁺ T cells infiltration. Previous literature has reported that TNBC cells exhibit inherent

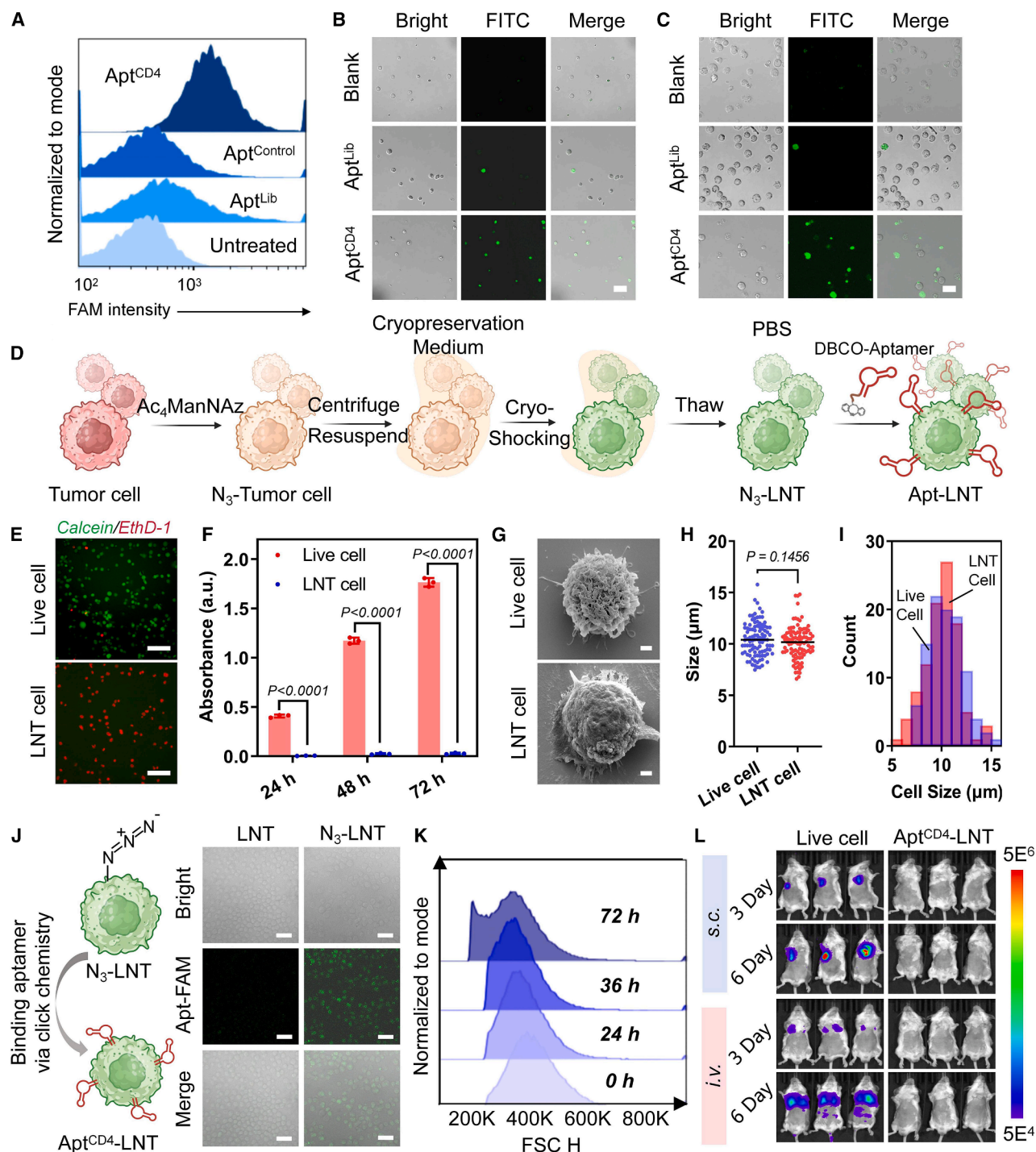


Figure 1. Design and characterization of Apt^{CD4}-LNTs

(A) Flow cytometry analysis of 6-Carboxyfluorescein (FAM)-labeled Apt^{CD4} binding to human CD4⁺ Jurkat cells.

(B and C) Confocal imaging characterization of aptamers' binding to mouse CD4⁺ T cells (B) and human CD4⁺ T cells (C). Scale bars, 20 μm.

(D) Schematic of the Apt-LNTs preparation workflow: metabolic glycan labeling, cryoinactivation in liquid nitrogen, and click-chemistry conjugation of DBCO-Apt^{CD4} to azide-engineered LNTs.

(E) Viability staining of live cells and LNTs. Calcein acetoxymethyl ester (Calcein-AM, green) marks live cells; Ethidium Homodimer-1 (EthD-1, red) marks dead cells; scale bar, 50 μm.

(F) Proliferation assay of live cells and LNTs (mean ± SD, *n* = 3).

(legend continued on next page)

tumor-homing and retention capacity.³³ This has been partially attributed to the expression of cell surface interleukin-6 receptor (IL-6R, acting as a chemoattractant for breast cancer cells) and Fascin-1 (mediating breast cancer cell invasion into tumors).³⁴ In line with these findings, we demonstrated the expression of IL-6R and Fascin-1 in 4T1 cells and Apt^{CD4}-LNTs (prepared from 4T1 tumor cells) by confocal imaging and flow cytometry (Figures S5A, S5C, and S5D).

To further confirm that the homing capacity of Apt^{CD4}-LNTs, we conducted *in vitro* transwell assays. The results showed that Apt^{CD4}-LNTs retained a strong homing capacity (Figures S6A and S6B). For *in vivo* validation, we established a subcutaneous (s.c.) tumor model of TNBC cell line 4T1 in mice and intravenously (i.v.) injected Apt^{CD4}-LNTs, labeled with Cy5.5 fluorescence for real-time tracking (Figure S7A). *In vivo* imaging showed comparable biodistribution of fluorescence in major organs across the LNT group, Apt^{control}-LNT group, and Apt^{CD4}-LNT group, with all exhibiting marked tumor accumulation (Figures 2B, S7B, and S7C). Furthermore, longitudinal imaging revealed that Apt^{CD4}-LNTs began to accumulate in tumors as early as 6 h post-injection and persisted for over 264 h, confirming sustained and specific tumor-homing (Figures S7D–S7F). These results established Apt^{CD4}-LNTs as an effective carrier for targeted delivery to TNBC tumors.

Having observed the homing capacity of Apt^{CD4}-LNTs, we then examined whether it selectively induces CD4⁺ T cell infiltration. *In vitro* simulation experiments demonstrated that Apt^{CD4}-LNTs can significantly enhance the capture efficiency of CD4⁺ T cells (Figure S8). Our CD4⁺ T cell dosing strategy used a consecutive triple-dose regimen to enhance *in vivo* infiltration efficacy and limited single administrations to $< 5 \times 10^6$ cells to minimize physiological stress (Figure 2C), referencing established CAR-T cell therapy protocols.³⁵ Flow cytometry analysis of mice tumor samples revealed that Apt^{CD4}-LNTs significantly increased intratumoral immune cells infiltration compared with both untreated and Apt^{control}-LNT groups, with a notable enrichment of CD4⁺ T cells (Figures 2D and S9). Multiplex immunofluorescence (mIHC) staining further confirmed that Apt^{CD4}-LNTs treatment specifically enhanced CD4⁺ T cells infiltration (Figures 2E, 2F, and S10C). Although a modest increase in CD4⁺ regulatory T cells (Tregs) was observed, they accounted for only a small fraction of the total CD4⁺ population and did not alter the dominant effector profile (Figures S10D and S10E). Notably, Apt^{CD4}-LNTs significantly enriched CD4⁺ T cells at the tumor margin (Figures 2G, S10A, and S10B), an area previously identified as a key function site for anti-tumor CD4⁺ T cell activity.⁹ In summary, Apt^{CD4}-LNTs effectively remodeled intratumoral CD4⁺ T infiltration and distribution, potentially improving the efficacy of CD4⁺ T cell-based therapy.

To determine whether this tumor-homing capacity is specific to 4T1-derived LNTs, we extended our analysis to two additional

murine TNBC cell lines, EMT6 and PyMT. We generated Apt^{CD4}-LNTs from each line (Apt^{CD4}-LNT^{EMT6} and Apt^{CD4}-LNT^{PyMT}) and evaluated their homing capacity *in vitro* and *in vivo*. In transwell assays, the Apt^{CD4}-LNT^{PyMT} group demonstrated stronger chemotactic behavior than Apt^{CD4}-LNT^{EMT6} (Figures S6C and S6D). *In vivo* imaging system (IVIS) imaging similarly showed preferential accumulation of Apt^{CD4}-LNTs^{PyMT} at tumor sites, while Apt^{CD4}-LNTs^{EMT6} exhibited weaker tumor localization (Figures S7G–S7I). To elucidate the molecular basis of these differences, we quantified IL-6R and Fascin-1 expression across the three cell lines. Both 4T1 and PyMT cells exhibited high expression levels of these markers, whereas EMT6 cells expressed significantly lower levels (Figures S5B–S5D). These data again established IL-6R and Fascin-1 as molecular markers of LNTs' homing capacity and underscore their roles in mediating targeted delivery.

Apt^{CD4}-LNTs suppressed tumor growth

To assess whether Apt^{CD4}-LNTs-induced CD4⁺ T cell infiltration impacts tumor growth, we divided 4T1 TNBC mice into four groups: untreated (H0), LNT (H1), Apt^{control}-LNT (H2), and Apt^{CD4}-LNT (H3) and followed the treatment protocol outlined in Figure 3A. Both fluorescence imaging and tumor volume measurements revealed that the Apt^{CD4}-LNT group significantly inhibited TNBC tumors compared with all control groups. While LNT (H1) and Apt^{control}-LNT (H2) groups showed modest inhibition of tumor cells, likely due to innate immune activation triggered by control nucleic acid components,³⁶ the Apt^{CD4}-LNT group (H3) exhibited the most pronounced suppression of tumor growth (Figures 3B and 3C). The Apt^{CD4}-LNT cohort exhibited 3.9-fold (vs. untreated), 3.4-fold (vs. LNT), and 2.4-fold (vs. Apt^{control}-LNT) greater tumor volume reduction on day 18 post-treatment (Figure 3D). Representative tumor images collected on day 9 also confirmed that Apt^{CD4}-LNTs markedly decreased 4T1 tumor size (Figure 3E). The survival curve (Figure 3F) also demonstrated that Apt^{CD4}-LNT group significantly prolonged survival in 4T1 tumor-bearing mice.

Given the high pulmonary metastatic propensity of 4T1 tumors, we evaluated metastatic dissemination 30 days post-subcutaneous implantation of 4T1-Luc cells. IVIS imaging of excised lung tissues revealed detectable metastases in 2 of 3 untreated mice, whereas no metastases were observed in the Apt^{CD4}-LNT group (Figure S11). Importantly, Apt^{CD4}-LNTs did not result in significant weight loss (Figure 3G) or major organs damage (Figure S12), indicating the safety of Apt^{CD4}-LNTs. These above results conclusively demonstrated the effective suppression of 4T1 TNBC tumors via Apt^{CD4}-LNTs.

Apt^{CD4}-LNTs selectively mobilized CD4⁺ T cells

To investigate whether the suppression of TNBC tumors is driven by Apt^{CD4}-LNTs-mediated CD4⁺ T cell mobilization, we performed single-cell RNA transcriptome sequencing (scRNA-seq)

(G) Scanning electron microscopy (SEM) images of live cells and LNTs. Scale bar, 1 μ m.

(H and I) Quantification of cell diameter for live cells and LNTs (mean $\pm$ SD, $n = 100$ cells).

(J) Confocal microscopy image of Cy5-labeled aptamer conjugated to N₃-LNTs. Scale bar, 50 μ m.

(K) Flow cytometry analysis of Apt^{CD4}-LNTs structural stability. Forward scatter height (FSC H) represents the cell sized particles.

(L) *In vivo* bioluminescence imaging of 4T1 tumor formation following subcutaneous (s.c.) or intravenous (i.v.) administration of live cells versus of Apt^{CD4}-LNTs.

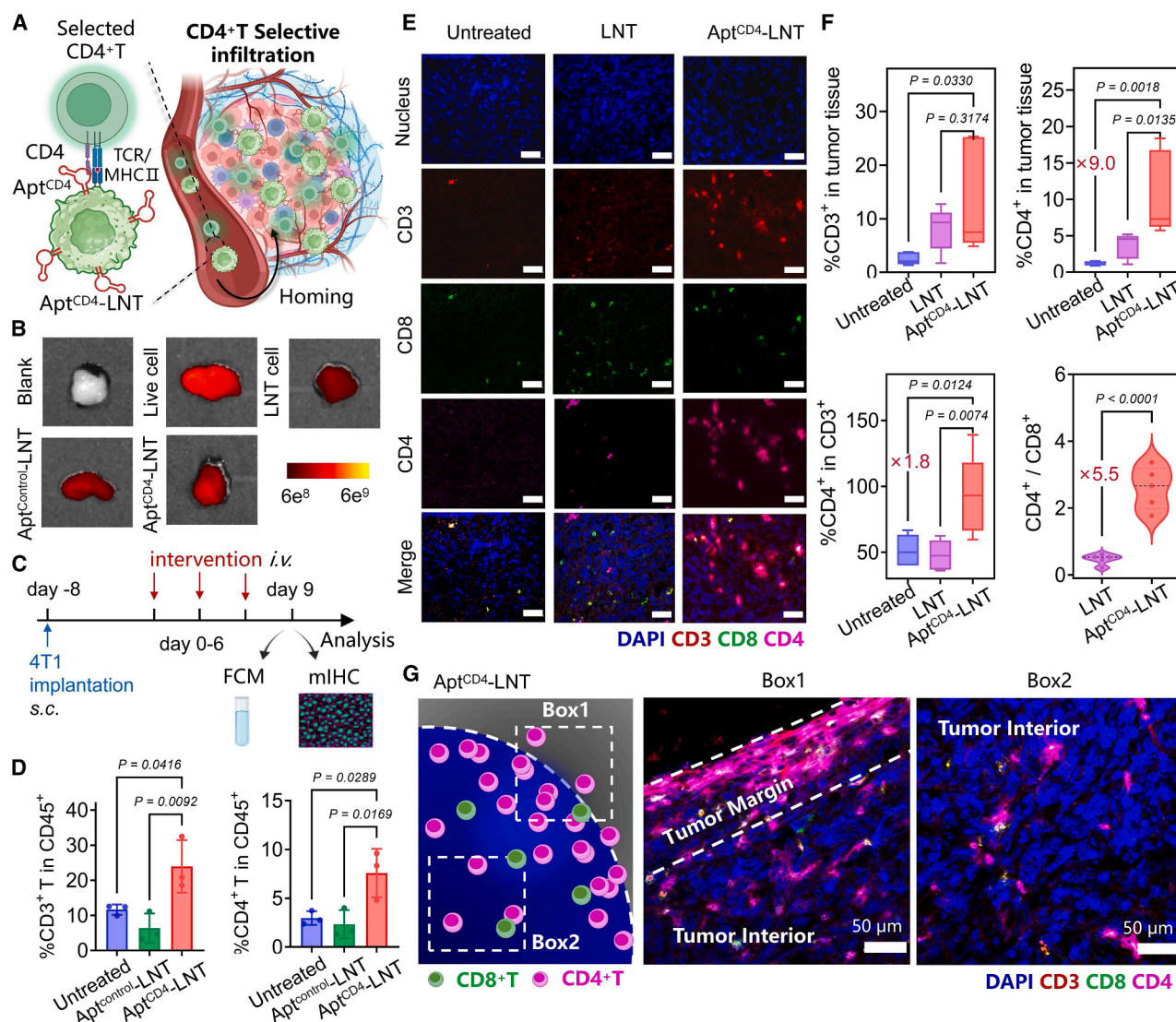


Figure 2. Selective CD4⁺ T cells infiltration by Apt^{CD4}-LNTs

(A) Schematic overview of Apt^{CD4}-LNTs-mediated CD4⁺ T cells recruitment to 4T1 tumors.

(B) Tumor tissue imaging showed the *in vivo* tumor-homing capacity of Cy5.5-labeled live cells, LNTs, Apt^{control}-LNTs, and Apt^{CD4}-LNTs.

(C) Experimental protocol referenced established dosing protocols for CAR-T cell therapies, with a single administration dose of fewer than 5×10^6 cells to minimize physiological stress.³⁵ For Apt^{CD4}-LNTs-mediated selective infiltration of immune cells, 5×10^6 , 4×10^6 , and 3×10^6 cells were injected intravenously on the 0th, 3rd, and 6th day, respectively.

(D) Flow cytometry (FCM) quantification of CD4⁺ T cell frequency across treatment groups. (mean $\pm$ SD, $n = 3$).

(E) Infiltration of CD3⁺ cells, CD4⁺ cells, and CD8⁺ cells in the tumor was analyzed by multiplex immunofluorescence (mIHC); scale bar, 50 μ m.

(F) Statistical analysis of CD4⁺ T cell density in tumor sections (mean $\pm$ SD, $n = 5$).

(G) Enlarged view of margin and interior of tumor section after Apt^{CD4}-LNTs intervention. Scale bar, 50 μ m.

on tumors harvested from untreated and Apt^{CD4}-LNT groups (Figure 4A). A total of 28,631 high-quality cells were successfully obtained, which were classified into distinct subpopulations (Figure S13), including tumor cells, stroma cells, T cells, B cells and DCs (B&DC cells), macrophages, and monocytes (Figure 4B). Results revealed that Apt^{CD4}-LNTs treatment reduced the proportion of tumor cells and increased immune cell infiltration (Figure 4C). Further analysis of T cell subpopulations revealed that Apt^{CD4}-LNTs treatment increased the percentage of CD4⁺ T cells

within the CD3⁺ T cell population by 2.6-fold, and the CD4⁺ T/CD8⁺ T ratio increased by 4.3-fold (Figures 4D and 4E). Although Treg cells infiltration increased in the tumor microenvironment (Figure S14A), their proportion remained relatively low (Figure S14B), consistent with mIHC results (Figures S10D–S10F). These findings further confirmed the specificity and effectiveness of Apt^{CD4}-LNTs in promoting CD4⁺ T cell infiltration.

In addition to promoting the selective infiltration of CD4⁺ T cells, Apt^{CD4}-LNTs treatment also increased the expression

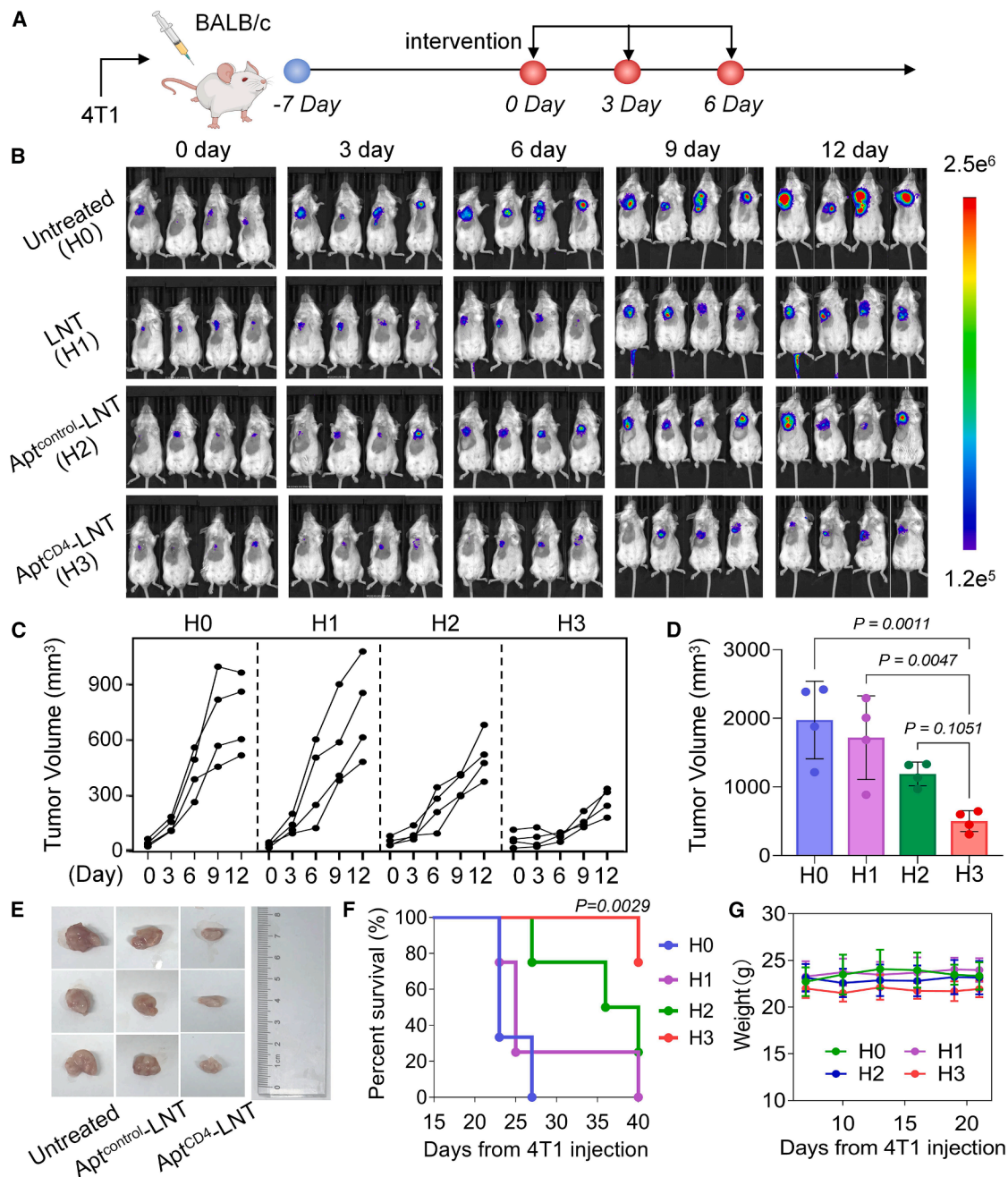


Figure 3. Apt^{CD4}-LNTs mediated TNBC tumor suppression

(A) Experimental protocol for Apt^{CD4}-LNT treatment, 5×10^6 , 4×10^6 , and 3×10^6 Apt^{CD4}-LNTs were injected by i.v. on the 0th, 3rd, and 6th day, respectively. (B) Representative bioluminescence images of 4T1-Luc tumors over time for untreated, LNT, Apt^{control}-LNT, and Apt^{CD4}-LNT groups. (C) Tumor growth curves of 4T1 bearing mice for indicated treatment groups. (D) Tumor size measure on day 18 after treatment (mean ± SD, n = 4). (E) Photographs of excised tumors on the 9th day after the first treatment for indicated treatment groups. (F) Survival curves of 4T1-bearing mice. Survival was statistically compared using a log-rank Mantel-Cox test (n = 4). (G) Body weight monitoring during treatment. Data represent mean ± SD (n = 4).

of several costimulatory genes, such as *CD69*, *ICOS*, *CD226*, *TNFRSF14*, *TNFRSF9*, and *CD28*, in CD4⁺ T cells (Figure 4F). Notably, the proportion of CD4⁺ T cells expressing *CD69*, an

early T cell activation marker, was significantly elevated (Figure 4G) suggesting that Apt^{CD4}-LNTs may selectively activate intratumoral CD4⁺ T cells (Figures 4G and 4H). These

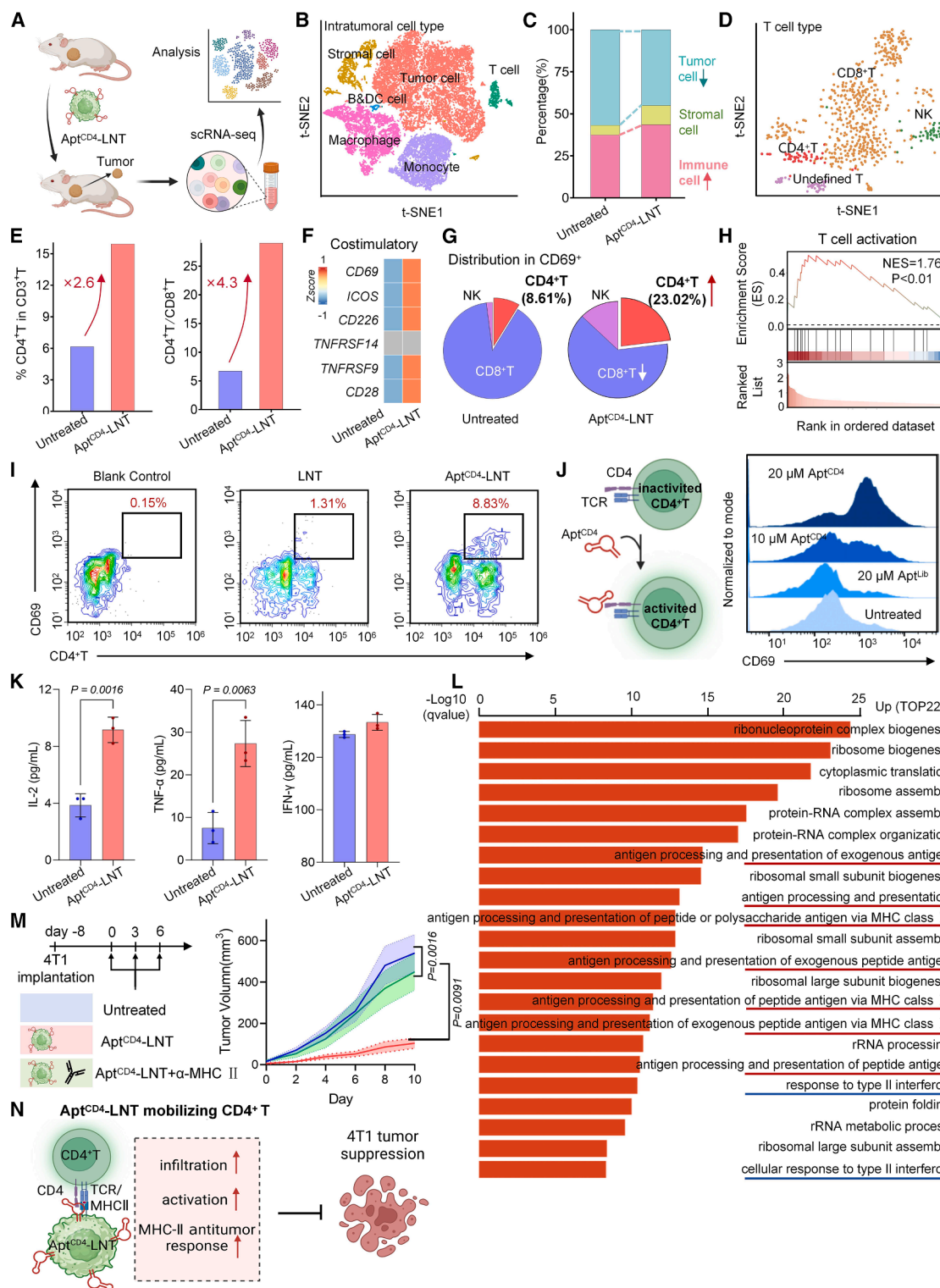


Figure 4. Apt^{CD4}-LNTs selectively mobilized CD4⁺ T cells

(A) Schematic diagram of Apt^{CD4}-LNTs mobilizing CD4⁺ T cells using single-cell analysis (scRNA-seq).

(B and D) t-distributed stochastic neighbor embedding (t-SNE) visualization of all cell clusters and T cell clusters, respectively.

(C) Relative percentages of tumor cells, stromal cells and immune cells for untreated group and Apt^{CD4}-LNT group in tumors.

(legend continued on next page)

findings were corroborated *in vitro*, where co-culture of Apt^{CD4}-LNTs with CD4⁺ T cells derived from C57BL/6 mice, BALB/c mice, and Jurkat T cells significantly upregulate CD69 expression and activate CD4⁺ T cells in a dose-dependent manner (Figures 4I, S15, and S16A–S16D). Given that the CD4 molecule acts as a costimulatory molecule for T cell receptor (TCR),^{37–39} we hypothesized that the interaction between the CD4 aptamer and the CD4 molecule plays a key role in the selective activation of CD4⁺ T cells (Figure 4J). Co-incubation experiments (Figures 4J, S16E, and S16F) demonstrated that compared with Apt^{Lib} and the untreated group, the Apt^{CD4} group had a higher proportion of CD69⁺ cells, confirming the ability of Apt^{CD4} to activate CD4⁺ T cells.

Functional activation was further supported by cytokine profiling: co-culture with Apt^{CD4}-LNTs significantly elevated interleukin (IL)-2, tumor necrosis factor (TNF)- α , and interferon (IFN)- γ , consistent with enhanced effector T cell activity (Figure 4K). Transcriptomic analysis further revealed that the Apt^{CD4}-LNT group upregulated genes associated with MHC II antigen processing and presentation, as well as response to IFN- γ (Figure 4L). To directly assess whether the anti-tumor response was MHC II-dependent, we blocked CD4⁺ T cell recognition of MHC II using anti-MHC II antibodies (α -MHC II) (Figure S17A). MHC II blockade abrogated 79.2% of Apt^{CD4}-LNTs anti-tumor efficacy, evidenced by tumor volume suppression (Figure 4M and S17B–S17D), directly establishing that Apt^{CD4}-LNTs can enhance MHC II-mediated anti-immune response. In summary, Apt^{CD4}-LNTs effectively mobilized CD4⁺ T cells, promoting their selective infiltration and activation. This, in turn, led to an enhanced MHC II-mediated anti-tumor immune response driven primarily by CD4⁺ T cells, resulting in TNBC tumor suppression (Figure 4N).

CD4⁺ T cells remodel the TNBC tumor microenvironment

To investigate whether CD4⁺ T cells exert a direct cytotoxic effect on tumors (Figure S18), we first confirmed the expression of MHC II on TNBC tumor cells (Figure S19). Further intercellular communication analysis showed that, although Apt^{CD4}-LNT group has enhanced overall cell-cell interactions, the interaction between CD4⁺ T cells and tumor cells was reduced (Figure 5A). Moreover, cytotoxicity-related gene expression in CD4⁺ T cells was downregulated following Apt^{CD4}-LNTs intervention (Figure 5B), suggesting that CD4⁺ T cell activation alone does not directly kill tumor cells.

However, single-cell analysis revealed an increase in the proportion of intratumoral B&DC cells, monocytes (Mono), and NK cells following Apt^{CD4}-LNTs treatment (Figures 5C and S19). These findings were further validated by mIHC and flow cytometry (FCM) (Figures 5D and S9), suggesting that CD4⁺ T cells may exert their anti-tumor effects through interactions with other immune cells (Figure S18). Further cell-cell communication analysis confirmed that Apt^{CD4}-LNTs treatment significantly enhanced interactions from CD4⁺ T cells to both B&DC cells and NK cells (Figure 5E). This enhanced interaction likely led to an increase in the proportion of cytotoxic NK cells (increased expression of tumor cell killing factor, *GZMA* and *PRF1*) and heightened B&DC cell activity, contributing to the anti-tumor response. Moreover, activated B and mature DC cells (identified by markers *CD80*, *CD86*, *CD40*, *CCR7*, and *IDO1* for mature DCs; *CD40*, *CD86*, *DHRS9*, and *EBI3* for activated B cells) were also found to be increased in the Apt^{CD4}-LNT group (Figure 5F). Together, these results indicate that Apt^{CD4}-LNTs enhanced interactions between CD4⁺ T cells and other anti-tumor immune cells, thereby improving the immune microenvironment and contributing to tumor suppression (Figure 5G).

DISCUSSION

The role of CD4⁺ T cells in anti-tumor immunotherapy is receiving increasing attention with various CD4⁺ T cell-based immunotherapy strategies now entering clinical studies.^{40,41} Despite these advancements, strategies to unleash the full potential of CD4⁺ T cells remain to be developed. The major limitation involves low immune infiltration of CD4⁺ T cell in tumors and the adverse anti-tumor effects of CD4⁺ T subtypes. In our work, we introduce Apt^{CD4}-LNTs as a novel platform that combines the innate tumor-homing capacity of cryo-shocked cancer cells (LNTs) with a CD4 aptamer to selectively mobilize, infiltrate, and activate CD4⁺ T cells within TNBC tumors. Our data demonstrate three key advances: (1) conjugation of a CD4-targeting aptamer enables specific capture and activation of CD4⁺ T cells; (2) the synergistic effect between the CD4-targeting aptamer and LNTs—derived from TNBC cells with inherent tumor microenvironment trafficking capability—enables selective intratumoral infiltration and activation of CD4⁺ T cells; and, (3) recruited CD4⁺ T cells remodel the intratumoral immune landscape—enhancing antigen presentation, costimulatory signaling, and recruitment of B cells, DCs, and NK cells—thereby suppressing tumor growth independent of CD8⁺ T cells.

(E) Percentages of CD4⁺ T cells in CD3⁺ T cells and ratio of CD4⁺ T cell sand CD8⁺ T cells for the untreated group and Apt^{CD4}-LNT group in tumors.

(F) Heatmap showing the expression of T cells' costimulatory genes in CD4⁺ T cells.

(G) Changes in the distribution of NK cells, CD8⁺ T cells, and CD4⁺ T cells in CD69⁺ cells induced by Apt^{CD4}-LNTs.

(H) Enrichment plot of the hallmark of T cell activation.

(I) *In vitro* CD4⁺ T cell activation via Apt^{CD4}-LNTs, as analyzed by flow cytometry, with CD69⁺ representing activated cells and CD3⁺ represents CD4⁺ T cells obtained by CD4⁺ T cell negative sorting from C57BL/6 mouse spleen.

(J) The effect of activated Apt^{CD4} on human CD4⁺ T cells, with CD69⁺ representing activated T cells.

(K) IL-2, TNF- α , and IFN- γ levels by Apt^{CD4}-LNTs treatment as indicated. Untreated: only CD4⁺ T cells; Apt^{CD4}-LNT: co-incubation of CD4⁺ T cells with Apt^{CD4}-LNTs with a ratio of 3:2 (mean $\pm$ SD, $n = 3$).

(L) Comparison of GO enrichment pathways analysis in intratumoral immune cells between the untreated group and Apt^{CD4}-LNT group. The top 22 pathways upregulated in Apt^{CD4}-LNT group are shown. Benjamini-Hochberg (BH) adjusted p value < 0.01.

(M) Protocol and tumor growth of mice treated with Apt^{CD4}-LNTs with or without anti-MHC II antibody blockade (mean $\pm$ SEM, $n = 6$).

(N) Schematic diagram of Apt^{CD4}-LNTs mobilizing CD4⁺ T cells to inhibit 4T1 tumors.

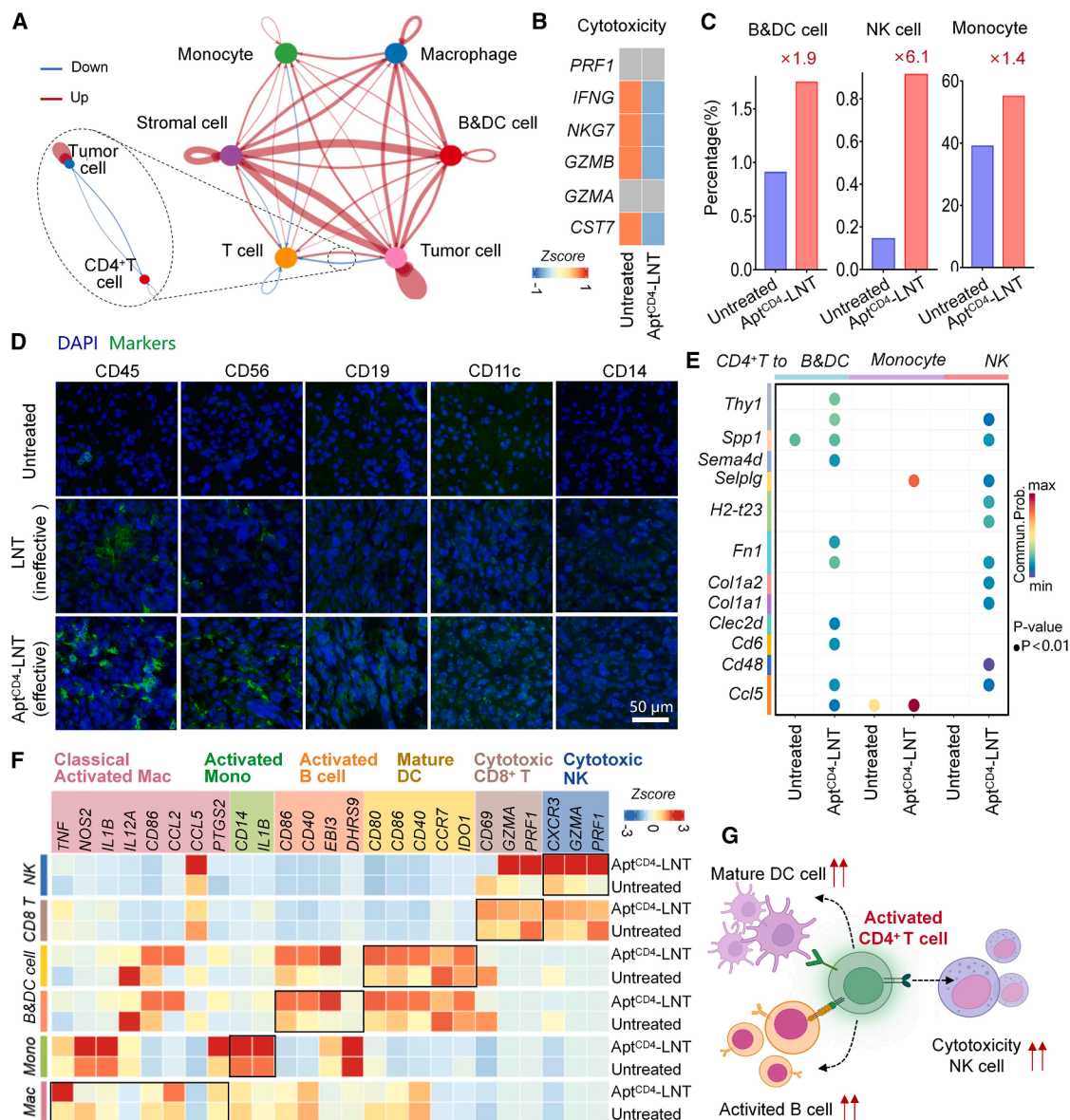


Figure 5. CD4⁺ T cells remodeling the tumor microenvironment for TNBC suppression

(A) By comparing between the untreated group and Apt^{CD4}-LNT group, a number of significant predicted intercellular interactions. The edge width is proportional to the indicated number of ligand-receptor pairs. Red represents upregulation after Apt^{CD4}-LNTs intervention, while blue represents downregulation after Apt^{CD4}-LNTs intervention.

(B) Heatmap showing the expression of cytotoxic genes in CD4⁺ T cells.

(C) Cell types with increased infiltration after Apt^{CD4}-LNTs intervention, showing percentage of B&DC, NK cells, and monocytes in immune cells.

(D) Infiltration of CD45⁺ cells (representing immune cells), CD56⁺ cells (representing NK cells), CD19⁺ cells (representing B cells), CD11c⁺ cells (representing DCs), and CD14⁺ cells (representing monocytes) in the tumor were analyzed by immunofluorescence; scale bar, 50 μm.

(E) Comparison of significantly increased ligand-receptor pairs between the untreated group and Apt^{CD4}-LNT group, from CD4⁺ T cells to B&DC, monocytes, and NK cells. The ligand genes on CD4⁺ T cells were displayed. Dot color reflects communication probabilities, and *p* values are computed from a one-sided permutation test.

(F) Expression patterns of related genes and cytokines in intratumoral classical activated macrophages (M1 Mac), activated monocytes (Mon), activated B cells and mature DCs (B&DC), cytotoxic CD8⁺ T cells, and cytotoxic NK cells.

(G) Anti-tumor mechanism of activated CD4⁺ T cells.

Our findings build on previous reports that TNBC cells—and by extension their LNTs derivatives—express IL-6R and Fascin-1 to home-into and adhere to tumor tissues.³⁴ By con-

firmed IL-6R and Fascin-1 expression in 4T1-derived LNTs and demonstrating their sustained tumor accumulation *in vivo*, we establish LNTs as effective, programmable delivery vehicles

for tumor-targeted immunomodulation. Importantly, homing capacity varied across different TNBC lines, correlating with differential IL-6R/Fascin-1 expression. This underscores the need to characterize or engineer LNTs homing capacity for each tumor type in future applications.

While CD8⁺ T cells have traditionally dominated cancer immunotherapy paradigms, our data reinforce the growing recognition of CD4⁺ T cells as potent effectors, especially in MHC I-deficient and immunosuppressive contexts. Apt^{CD4}-LNTs selectively increased the intratumoral CD4⁺/CD8⁺ ratio and upregulated costimulatory and activation markers (CD69, ICOS, and CD28, etc.), driving an MHC II-dependent anti-tumor response. Further, scRNA-seq analysis revealed that CD4⁺ T cells did not directly kill tumor cells but orchestrated a broader tumor microenvironment remodeling, including increased infiltration of activated B cells, mature DCs, and cytotoxic NK cells, along with the production of related cytokines, effectively suppressing TNBC tumor growth in a CD8⁺ T cell-independent manner. These results underscore the critical role of selective immune cell mobilization in tumor immunotherapy, offering avenues for CD4⁺ T cell-based strategies in TNBC treatment. Our findings align with recent clinical advancements, such as MAGE-A3-specific CD4⁺ T cell therapy for melanoma,^{40,41} and CD4⁺ CAR-T,^{17,42} which have been demonstrated superior efficacy of CD4⁺ T cells compared with CD8⁺ T cells.

Despite its promise, the Apt^{CD4}-LNTs platform has limitations. First, tumor-homing capacity and anti-tumor efficacy is not universal across all TNBC models (Figure S20), highlighting the need to elucidate or engineer surface determinants (e.g., IL-6R, Fascin-1) to achieve consistent targeting. Second, while Apt^{CD4}-LNTs effectively mobilized CD4⁺ T cells, the modest increase in Tregs warrants further study to ensure that immunosuppressive subsets do not offset therapeutic benefits. Finally, long-term efficacy and potential immunogenicity of repeated Apt^{CD4}-LNTs dosing remain to be assessed.

Looking forward, optimizing LNTs' homing by screening additional cell lines, or engineering synthetic membrane vesicles may broaden applicability across diverse tumor types. Combining Apt^{CD4}-LNTs with checkpoint inhibitors or adoptive CD8⁺ T cell therapies could synergize helper and cytotoxic responses. Moreover, expanding the aptamer repertoire to other immune targets (e.g., CD137 and OX40) may enable fine-tuning of intratumoral signaling. Overall, Apt^{CD4}-LNTs serve as a versatile, modular platform that harnesses CD4⁺ T cell biology to reshape the tumor microenvironment, opening new avenues for TNBC and potentially other "cold" tumors refractory to conventional immunotherapies.

METHODS

Materials, cells, and animals

N-azidoacetylmannosamine (Ac₄ManNAz) reagent was purchased from MedChemExpress. Cell Counting Kit (CCK-8) was obtained from Lablead. Calcein-AM was obtained from Beyotime. EthD-1 was obtained from Thermo Fisher. All oligonucleotides were synthesized and modified by Sangon Biotech Company and purified using reversed phase high-performance liquid chromatography (HPLC). Mouse and human CD4⁺ T cell isolation kits were obtained from BioLegend.

TNBC cell lines (4T1 cells, EMT6 cell, and PyMT cell line, Py8119 cells) and TNBC cells expressing luciferase (4T1-Luc cells) were purchased from American type culture collection (ATCC) and cultured in RPMI 1640 medium and DMEM medium, respectively, supplemented with 10% FBS and penicillin-streptomycin (P/S, 100 IU/mL, Gibco) at 37°C in a humidified incubator containing 5% CO₂.

Female BALB/c mice and C57BL/6J mice, aged 8- to 10-weeks, were purchased from GemPharmatech. Animal experiments were conducted following protocols approved by the Hangzhou Institute of Medicine, Chinese Academy of Sciences, China. The approval number is AP2024-12-0382. Mice were sacrificed when tumors exceeded 1,500 mm³ mean size or when mice showed signs of sickness in accordance with the local ethical regulations. All animal experiments were performed in groups of 4 to 6 mice and repeated independently at least twice.

Primary mouse CD4⁺ T cells were isolated from C57BL/6J or BALB/c mice, while human CD4⁺ T cells originated from the Jurkat cell line. For primary cell isolation, 8- to 10-week-old mice were euthanized and spleens harvested; spleen tissue was washed and mechanically dissociated in fresh dulbecco's phosphate-buffered saline (DPBS), then centrifuged. Cells were treated with 2 mL red blood cell lysis buffer for 3–5 min, lysing was terminated with 2 mL DPBS, followed by centrifugation, resuspension, and counting. CD4⁺ T cells were isolated via CD4⁺ T cell Negative Selection Kit (BioLegend) per the manufacturer's protocol. Antibody and magnetic bead incubation enabled magnetic retention of non-target cells, allowing collection of CD4⁺ T cells from the supernatant after centrifugation. Finally, CD4⁺ T cells were cultured in RPMI 1640 medium supplemented with 10% FBS, penicillin-streptomycin, and 0.1% 2-mercaptoethanol at 37°C in a humidified 5% CO₂ incubator.

Characterization of aptamers

6-Carboxyfluorescein (FAM)-labeled aptamers were purchased from Sangon Biotech Company. Washing buffer was prepared by supplementing DPBS with 4.5 g/L glucose, 1.2 mM CaCl₂, and 5 mM MgCl₂. Binding buffer was prepared by adding 0.4 mg/mL yeast tRNA and 1 mg/mL BSA to the washing buffer. Prior to binding, FAM-labeled aptamers were heated to 95°C for 15 min for complete denaturation and then cooled to 4°C. Cells were washed three times with the washing buffer and subsequently incubated with 250 nM aptamers (in binding buffer) for 90 min at room temperature. After washing with the washing buffer, cells were collected in DPBS for flow cytometry analysis.

Aptamer stability testing

Aptamers were dissolved in double distilled water (ddH₂O) to 200 μM. Aptamers were diluted to 5 μM in DPBS containing 20% FBS and incubated at 37°C for 0, 6, 12, 24, 48, 72, and 96 h. After heating the aptamers to 95°C for 15 min to achieve complete denaturation, their aptamer stability was analyzed using SDS-PAGE, with a low-molecular-weight DNA marker (20–120 nt) serving as a reference marker.

CD4 protein binding assay

0.005 μmol DBCO-modified aptamers were incubated with 0.5 mg azide-functionalized magnetic beads (PuriMag Biotech) at 25°C overnight. After magnetic separation and washing, aptamer-conjugated beads were annealed at 95°C

for 10 min. Subsequently, beads were co-incubated with 20 $\mu\text{g/mL}$ fluorescein isothiocyanate (FITC)-labeled CD4 protein (ACROBiosystems) in binding buffer for 90 min. After magnetic separation and washing, CD4 protein binding was quantified via flow cytometry.

Preparation and characterization of Apt-LNTs

Over a 48-h period, 4T1 cells were cultured in culture medium with 100 μM Ac₄ManNAz to obtain N₃-tumor cells, which were harvested and dispersed in the cryopreservation medium with a density of $3 \times 10^6/\text{mL}$. Cell medium was subsequently submerged in liquid nitrogen (F group) or gradient freezing (S group) for more than 24 h. Before use, cells were thawed in a 37°C water bath, pelleted at $1,000 \times g$ for 5 min, and washed once with DPBS, followed by dispersal in DPBS with $1 \times 10^6/\text{mL}$ to obtain N₃-LNTs. After annealing at 95°C, DBCO-modified aptamers were added at a ratio of 0.05 nmol per 1×10^5 N₃-LNTs and incubated for 1.5 h. Apt-LNTs were obtained by centrifugation and washing. The modification of aptamers (FAM-labeled) was analyzed by laser scanning confocal microscopy (A1 HD25, NIKON). Other TNBC cell-derived Apt-LNTs were prepared using their respective live cells.

Viability analysis

Apt-LNTs were treated with Calcein-AM and EthD-1 for 30 min at 37°C, followed by analysis by fluorescence microscopy (CKX53, Olympus). The cell size data presented in Figures 1H and 1I were derived from fluorescence imaging datasets. Moreover, a CCK-8 assay was performed to evaluate the viability of LNTs by microplate reader (Spark, Tecan).

Cell structure analysis

LNTs were fixed in 4% glutaraldehyde for 15 min. Afterward, a series of gradient ethanol (30%, 50%, 70%, 80%, and 90%) were used to dehydrate for 10 min, and 100% ethanol was utilized to wash twice for 10 min. The dehydrated LNTs then were dropped on an Au wafer and further analyzed by scanning electron microscopy (JSM-IT800, JEOL).

Modified efficiency analysis

DBCO-Apt^{CD4}-FAM was selected to establish an aptamer concentration-FAM fluorescence intensity standard curve using a microplate reader. Apt-LNTs were prepared via the aforementioned methods, then seeded into 96-well plates at 2.5×10^5 or 5×10^5 cells per well and subjected to aptamer fluorescence intensity measurement via microplate reader. The Apt^{CD4} modification rate per LNT cell was calculated using the following formula (Equations 1 and 2).

$$\text{Apt}^{\text{CD4}} \text{ per LNT cell (nmol)} = \frac{\text{Apt}^{\text{CD4}} \text{ per well (nmol)}}{\text{number of LNT cells per well}} \quad (\text{Equation 1})$$

$$N \text{ (number)} = \text{Apt}^{\text{CD4}} \text{ per LNT cell (mol)} \times N_A \text{ (} 6.02 \times 10^{23} \text{)} \quad (\text{Equation 2})$$

Apt-LNTs structural stability

Apt-LNTs were dispersed and maintained in DPBS at 37°C. Flow cytometry detection was performed at different times, and the size of Apt-LNTs were characterized by forward scatter height (FSC H) values.

Stability of aptamers conjugated to LNTs (FAM-labeled aptamers)

Apt-LNTs were dispersed in DPBS containing 20% FBS, incubated at 37°C for 0, 6, 24, and 48 h, centrifuged, and washed to collect the Apt-LNTs and subsequently analyzed by flow cytometry to evaluate aptamer stability based on LNTs surface fluorescence intensity.

Tumorigenicity studies were carried out using 4T1 cells expressing luciferase, which were used to prepare LNTs, and live tumor cells and LNTs were transplanted into mice by the s.c. (1×10^5 cells) and i.v. (3×10^6 cells) routes, respectively. Tumorigenicity on day 3 and day 6 was characterized using IVIS (Lumina Series III, PE).

For CD4 protein binding testing, 5×10^5 Apt^{CD4}-LNTs were resuspended in 250 μL binding buffer, followed by addition of 20 $\mu\text{g/mL}$ CD4 protein. After co-incubation for 90 min at room temperature, Apt^{CD4}-LNTs were collected via centrifugation and washing. FITC fluorescence intensity of Apt^{CD4}-LNTs was then analyzed by flow cytometry to evaluate CD4 protein binding capacity.

In vivo oligonucleotide metabolism analysis

2.5 nmol DBCO-Apt^{CD4}-Cy5 was intravenously injected into female BALB/c mice (aged 8–10 weeks), with aptamer metabolic kinetics monitored using IVIS. For Apt^{CD4}-LNTs metabolism studies, Apt^{CD4}-LNTs were prepared using DBCO-Apt^{CD4}-Cy5; subsequently, 5×10^6 Apt^{CD4}-LNTs were i.v. injected into female BALB/c mice (aged 8–10 weeks), and aptamer metabolic kinetics were similarly tracked using IVIS.

Self-homing of Apt-LNTs

For self-homing analysis, Cy5.5-N-Hydroxysuccinimide (NHS) was mixed with live 4T1 cells, LNTs, Apt^{control}-LNTs, and Apt^{CD4}-LNTs. 1×10^5 4T1 cells were transplanted in female BALB/c mice (aged 8–10 weeks). Cy5.5-labeled live 4T1 cells, LNTs, Apt^{control}-LNTs, and Apt^{CD4}-LNTs were injected into tumor-bearing mice by i.v., and self-homing was observed by IVIS (Lumina Series III, PE). For EMT6-derived Apt^{CD4}-LNTs, following the aforementioned procedures, 2×10^5 EMT6 cells were transplanted into female BALB/c mice (aged 8 to 10 weeks), with Apt^{CD4}-LNTs prepared using live EMT6 cells. For PyMT-derived Apt^{CD4}-LNTs, under identical conditions, 2×10^5 PyMT cells (Py8119) were transplanted into female C57BL/6J mice (aged 8–10 weeks), and Apt^{CD4}-LNTs were prepared using live PyMT cells.

Transwell experiment

1×10^5 4T1 cells were cultured in the lower chamber. For the control group, the lower chamber did not cultivate 4T1 cells. After 12 h, 2×10^5 Apt^{CD4}-LNTs (FITC-labeled aptamer, LNTs derived from 4T1) were cultured in the upper chamber. After co-cultivation for 24 h, Apt^{CD4}-LNTs in the lower chamber were analyzed by fluorescence microscopy. Other TNBC cell-derived Apt^{CD4}-LNTs followed the aforementioned procedures with the following modifications: the tumor cells were replaced with EMT6 and PyMT (Py8119) cells respectively, and Apt^{CD4}-LNTs were correspondingly prepared using live EMT6 and PyMT (Py8119) cells for subsequent experiments.

Flow analysis for immune cell infiltration

1×10^5 4T1-Luc cells were transplanted in female BALB/c mice (aged 8 to 10 weeks). Tumor-bearing mice were treated three times with Apt^{control}-LNTs or Apt^{CD4}-LNTs at a dose of 5×10^6 cells, 4×10^6 cells, and 3×10^6 cells, sequentially. Mice without any treatment served as control. After receiving different Apt-LNTs administration, the mice were sacrificed on day 9, and intratumoral CD45⁺ cells, CD3⁺ T cells, CD4⁺ T cells, CD8⁺ T cells, and NK cells in the TNBC tumor were analyzed using flow cytometry (CytoFLEX LX, Beckman Coulter). Markers and gating were shown in Table S2.

Immunofluorescence

After receiving Apt^{CD4}-LNTs administration on day 9, the mice were sacrificed, and tumor tissues were fixed and made into frozen sections. 3% BSA was used to block the non-specific binding. CD3 was labeled with CD3 primary antibody (1:500, GB12014, Servicebio) and marked with Cy3-conjugated secondary antibody (1:300, GB21301, Servicebio). CD4 was labeled with CD4 primary antibody (1:4,000, GB15064, Servicebio) and marked with horseradish peroxidase (HRP)-conjugated secondary antibody (1:500, GB23303, Servicebio). CD8 was labeled with CD8 primary antibody (1:500, GB15068, Servicebio) and marked with Alexa Fluor 488-conjugated secondary antibody (1:400, GB25303, Servicebio). Immune cells were labeled with CD45 primary antibody (1:500, GB113886, Servicebio), NK cells were labeled with CD56 primary antibody (1:500, GB112671, Servicebio), B cells were labeled with CD19 primary antibody (1:500, GB11061-1, Servicebio), DCs were labeled with CD11c primary antibody (1:300, GB11059, Servicebio), and monocyte cells were labeled with CD14 primary antibody (1:1,000, GB11254, Servicebio), and all were marked with Alexa Fluor 488-conjugated secondary antibody (1:400, GB25303, Servicebio). Nuclei were stained with DAPI. Treg cells were marked via CD4 and forkhead box protein p3 (Foxp3). CD4 was labeled with CD4 primary antibody and marked with HRP-conjugated secondary antibody. Foxp3 was labeled with Foxp3 primary antibody (1:300, GB112325, Servicebio) and marked with Cy3-conjugated secondary antibody.

Live cells were stained following adherence in confocal dishes, while Apt^{CD4}-LNTs were seeded into confocal dishes after staining. IL-6R was labeled with IL-6R primary antibody and marked with FITC-conjugated secondary antibody. Fascin-1 was labeled with primary antibody (SDT-759-13, STARTER) and marked with FITC-conjugated secondary antibody.

All the stained slides and cells were observed by a pathological slice scanning system (VS200, Olympus Corporation) or laser scanning confocal microscopy, and the fluorescence intensity was analyzed with ImageJ software.

In vivo Apt^{CD4}-LNTs treatment of 4T1 cells

To evaluate the effect of Apt^{CD4}-LNTs treatment, BALB/c mice (8–10 weeks old) were s.c. injected with 4T1-Luc cells (1×10^5 cells). Experimental protocol referenced established dosing protocols for CAR-T cell therapies, with a single administration dose of fewer than 5×10^6 cells to minimize physiological stress.³⁵ On days 0, 3, and 6 post-inoculation, mice received sequential tail-vein injections of Apt^{CD4}-LNTs at dose of 5×10^6 cells, 4×10^6 cells, and

3×10^6 cells, respectively (control groups received equivalent doses of LNTs or Apt^{control}-LNTs).

To observe the growth kinetics of tumors, the bioluminescence signals of mice were captured in every group every 3 days by IVIS. Tumor volume and mouse body weight in each treatment group were measured by vernier caliper and digital bathroom scale, respectively. Tumor sizes were detected by vernier caliper, and the volume was calculated as $V = A \times B^2 \times 0.52$, where A and B are the longer and shorter diameter (mm) of the tumor. To evaluate the potential side effects, after Apt-LNTs administration, the main organs were sliced for staining with hematoxylin and eosin (H&E). To detect lung metastasis, on the 30th day after 4T1-Luc tumor implantation, mice were sacrificed, lungs were extracted, and IVIS imaging was performed.

Supplementary treatment of EMT6 and PyMT

For EMT6 treatment, BALB/c mice (aged 8–10 weeks) were s.c. injected with EMT6 (2×10^5 cells). Then, Apt^{CD4}-LNTs (derived from EMT6) were i.v. injected with a dose of 5×10^6 cells, 4×10^6 cells, and 3×10^6 cells, sequentially. The PyMT therapeutic experiment followed procedures analogous to those described above, utilizing C57BL/6J mice (aged 8–10 weeks) as the model system, with Apt^{CD4}-LNTs prepared using live Py8119 cells. Tumor volume and mouse body weight in each treatment group were measured by vernier caliper and digital bathroom scale, respectively. Tumor sizes were detected by vernier caliper, and the volume was calculated as $V = A \times B^2 \times 0.52$, where A and B are the longer and shorter diameter (mm) of the tumor.

Kaplan-Meier survival curve analysis

The Kaplan-Meier Plotter database (Database: <http://kmplot.com/analysis/>) was used to evaluate the prognostic performance of different CD4 expression levels in human TNBCs. In this database, we identified the association between CD4 expression and overall survival (OS) in TNBC. Hazard ratios (HRs) with corresponding 95% confidence intervals (CIs) and log-rank *p* values were calculated.

scRNA-seq data processing

After receiving Apt^{CD4}-LNTs administration, sequentially, mice were sacrificed at 9 days, and tumors of three mice per group were pooled together as one sample for scRNA-seq following standard protocols.^{43–45} The Cell Ranger toolkit (v.7.2.0) provided by 10x Genomics was applied to aggregate raw data, filter low-quality reads, align reads to the mouse reference genome (GRCm39), assign cell barcodes, and generate the unique molecular identifier (UMI) matrix. The R-based toolkit, Seurat (v.4.4.0) was used for analyzing the scRNA-seq data. We quantified the number of genes and UMI count for each cell and kept high-quality cells with thresholds of 200–10,000 genes and less than 10% mitochondrial gene counts to ensure that most heterogeneous cell types were included for downstream analyses. The normalized expression matrix was calculated based on the raw UMI counts after normalizing total counts per cell (library size) and then scaled by $1e^4$ and logarithmically transformed.

Single-cell transcriptome data of immune cells were determined using the FindMarkers function in Seurat (v.4.4.0) with thresholds set to negative infinity. Significantly differentially expressed genes (DEGs) were selected by setting the threshold to $p < 0.05$ and absolute value of $\log_2\text{fc} > 0.25$.

Functional analysis was performed using R package clusterProfiler (v.4.10.0) in the R package. Gene Ontology (GO) mouse gene sets were assessed for 4T1 mouse model single-cell sequencing. Default unadjusted p values were used and displayed in figures.

In vitro mouse T cell activation

CD4⁺ T cells in the spleen (C57BL/6J mice or BALB/c) were separated by using mouse CD4⁺ T cell isolation kit (BioLegend). Then, the CD4⁺ T cells were co-incubated with Apt^{CD4}-LNTs with a ratio of 3:2. Using flow cytometry and mCD69⁺- and mCD3⁺-representing activated cells and CD4⁺ T cells, respectively, the activation efficiency of CD4⁺ T cells was analyzed. Markers and gating of primary cells were shown in Table S2.

Human T cells experiment

Jurkat cells (reported CD4-positive³¹) served as the human CD4⁺ T cell model. To evaluate Apt^{CD4} binding specificity, Ramos cells were selected as the CD4-negative control. Prior to binding, FAM-labeled aptamers were denatured at 95°C for 15 min and cooled to 4°C. Both cell types were washed with washing buffer, incubated with 250 nM aptamers in binding buffer for 90 min at room temperature, washed with washing buffer, and resuspended in DPBS for flow cytometry. For T cell activation analysis, 2×10^5 Jurkat cells were seeded in 96-well plates, treated with Apt^{CD4}-LNTs (3×10^5 per well), DBCO-Apt^{CD4}, or Apt^{Lib} and co-incubated for 24 h. After co-incubation for 24 h, CD4⁺ T cell activation efficiency was analyzed by flow cytometry. For the aptamer-treated group, T cell activation was assessed via anti-human CD69 (anti-hCD69) antibody staining. For the Apt^{CD4}-LNT group, co-staining with anti-hCD69 and anti-hCD3 antibodies was performed, thus enabling exclusion of Apt^{CD4}-LNTs during flow cytometry gating.

α -MHC II blockade therapy

Apt^{CD4}-LNTs administration followed the method described above. The Apt^{CD4}-LNT + α -MHC II combination group received sequential i.v. injections of 500, 400, and 300 μg anti-mouse MHC class II antibody (α -MHC II, clone Y-3P, Bio X Cell) 6 h prior to each corresponding Apt^{CD4}-LNTs injection.

Statistical analysis

Data were expressed as mean values of at least three independent experiments in each group $\pm$ standard deviations (SD) or standard error of the mean (SEM). The data were compared by Student's t test between two groups and ordinary one-way analysis of variance (ANOVA) for three or more groups. Survival curves were analyzed using the log-rank (Mantel-Cox) test.

RESOURCE AVAILABILITY

Lead contact

Further information and requests should be directed to and will be fulfilled by the lead contact, Weihong Tan (tan@hnu.edu.cn).

Materials availability

Materials used in this study are available upon request. No unique reagents were generated for this study. The composition and preparation of Apt^{CD4}-LNTs developed in this study is described in this paper.

Data and code availability

Processed single-cell RNA-seq data of this study can be obtained from Gene Expression Omnibus (GEO), GEO: GSE290903. The association between CD4 expression and clinical prognosis of TNBC patients can be obtained from Kaplan-Meier Plotter database (Database: <http://kmplot.com/analysis/>). The raw FASTQ files in this study will be provided for scientific research upon requirements. Code can be obtained with the DOI number, Zenodo: <https://doi.org/10.5281/zenodo.16740264>.

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AUTHOR CONTRIBUTIONS

Conceptualization, H.T., W.T., and Q.W.; methodology, Y.X., H.T., Q.W., S.G., L.F., L.G., K.Z., and H.L.; investigation, H.T., Q.W., T.F., and W.T.; visualization, H.T., Q.W., Y.X., L.F., and L.G.; supervision, W.T. and Q.W.; writing, H.T., W.T., Q.W., and Y.X.

DECLARATION OF INTERESTS

The authors have filed a patent application on aspects of this work.

SUPPLEMENTAL INFORMATION

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REFERENCES

- Kelly, P.N. (2018). The cancer immunotherapy revolution. *Science* 359, 1344–1345. <https://doi.org/10.1126/science.359.6382.1344>.
- Del Paggio, J.C.D. (2018). Immunotherapy: Cancer immunotherapy and the value of cure. *Nat. Rev. Clin. Oncol.* 15, 268–270. <https://doi.org/10.1038/nrclinonc.2018.27>.
- Giles, J.R., Globig, A.-M., Kaech, S.M., and Wherry, E.J. (2023). CD8⁺ T cells in the cancer-immunity cycle. *Immunity* 56, 2231–2253. <https://doi.org/10.1016/j.immuni.2023.09.005>.
- Puig-Saus, C., Sennino, B., Peng, S., Wang, C.L., Pan, Z., Yuen, B., Purandare, B., An, D., Quach, B.B., Nguyen, D., et al. (2023).

- Neoantigen-targeted CD8⁺ T cell responses with PD-1 blockade therapy. *Nature* 615, 697–704. <https://doi.org/10.1038/s41586-023-05787-1>.
5. Rudloff, M.W., Zumbo, P., Favret, N.R., Roetman, J.J., Detrés Román, C.R.D., Erwin, M.M., Murray, K.A., Jonnakuti, S.T., Dündar, F., Betel, D., et al. (2023). Hallmarks of CD8⁺ T cell dysfunction are established within hours of tumor antigen encounter before cell division. *Nat. Immunol.* 24, 1527–1539. <https://doi.org/10.1038/s41590-023-01578-y>.
 6. Chow, A., Perica, K., Klebanoff, C.A., and Wolchok, J.D. (2022). Clinical implications of T cell exhaustion for cancer immunotherapy. *Nat. Rev. Clin. Oncol.* 19, 775–790. <https://doi.org/10.1038/s41571-022-00689-z>.
 7. McGranahan, N., Rosenthal, R., Hiley, C.T., Rowan, A.J., Watkins, T.B.K., Wilson, G.A., Birkbak, N.J., Veeriah, S., Van Loo, P.V., Herrero, J., et al. (2017). Allele-specific HLA loss and immune escape in lung cancer evolution. *Cell* 171, 1259–1271.e11. <https://doi.org/10.1016/j.cell.2017.10.001>.
 8. Beck, J.D., Diken, M., Suchan, M., Streuber, M., Diken, E., Kolb, L., Allnoch, L., Vascotto, F., Peters, D., Beißert, T., et al. (2024). Long-lasting mRNA-encoded interleukin-2 restores CD8⁺ T cell neoantigen immunity in MHC class I-deficient cancers. *Cancer Cell* 42, 568–582.e11. <https://doi.org/10.1016/j.ccell.2024.02.013>.
 9. Kruse, B., Buzzai, A.C., Shridhar, N., Braun, A.D., Gellert, S., Knauth, K., Pozniak, J., Peters, J., Dittmann, P., Mengoni, M., et al. (2023). CD4⁺ T cell-induced inflammatory cell death controls immune-evasive tumours. *Nature* 618, 1033–1040. <https://doi.org/10.1038/s41586-023-06199-x>.
 10. Speiser, D.E., Chijioke, O., Schaeuble, K., and Münz, C. (2023). CD4⁺ T cells in cancer. *Nat. Cancer* 4, 317–329. <https://doi.org/10.1038/s43018-023-00521-2>.
 11. Bawden, E.G., Wagner, T., Schröder, J., Effern, M., Hinze, D., Newland, L., Attrill, G.H., Lee, A.R., Engel, S., Freestone, D., et al. (2024). CD4⁺ T cell immunity against cutaneous melanoma encompasses multifaceted MHC II-dependent responses. *Sci. Immunol.* 9, eadi9517. <https://doi.org/10.1126/sciimmunol.adi9517>.
 12. Borst, J., Ahrends, T., Băbala, N., Melief, C.J.M., and Kastenmüller, W. (2018). CD4⁺ T cell help in cancer immunology and immunotherapy. *Nat. Rev. Immunol.* 18, 635–647. <https://doi.org/10.1038/s41577-018-0044-0>.
 13. Laidlaw, B.J., Craft, J.E., and Kaech, S.M. (2016). The multifaceted role of CD4⁺ T cells in CD8⁺ T cell memory. *Nat. Rev. Immunol.* 16, 102–111. <https://doi.org/10.1038/nri.2015.10>.
 14. Okamura, H., Tsutsi, H., Komatsu, T., Yutsudo, M., Hakura, A., Tanimoto, T., Torigoe, K., Okura, T., Nukada, Y., and Hattori, K. (1995). Cloning of a new cytokine that induces IFN- γ production by T cells. *Nature* 378, 88–91. <https://doi.org/10.1038/378088a0>.
 15. Hasegawa, T., Oka, T., Son, H.G., Oliver-García, V.S., Azin, M., Eisenhaure, T.M., Lieb, D.J., Hacohen, N., and Demehri, S. (2023). Cytotoxic CD4⁺ T cells eliminate senescent cells by targeting cytomegalovirus antigen. *Cell* 186, 1417–1431.e20. <https://doi.org/10.1016/j.cell.2023.02.033>.
 16. Oh, D.Y., and Fong, L. (2021). Cytotoxic CD4⁺ T cells in cancer: expanding the immune effector toolbox. *Immunity* 54, 2701–2711. <https://doi.org/10.1016/j.immuni.2021.11.015>.
 17. Wang, D., Aguilar, B., Starr, R., Alizadeh, D., Brito, A., Sarkissian, A., Ostberg, J.R., Forman, S.J., and Brown, C.E. (2018). Glioblastoma-targeted CD4⁺ CAR T cells mediate superior antitumor activity. *JCI Insight* 3, e99048. <https://doi.org/10.1172/jci.insight.99048>.
 18. Seung, E., Xing, Z., Wu, L., Rao, E., Cortez-Retamozo, V., Ospina, B., Chen, L., Beil, C., Song, Z., Zhang, B., et al. (2022). A trispecific antibody targeting HER2 and T cells inhibits breast cancer growth via CD4 cells. *Nature* 603, 328–334. <https://doi.org/10.1038/s41586-022-04439-0>.
 19. Spitzer, M.H., Carmi, Y., Reticker-Flynn, N.E., Kwek, S.S., Madhiredy, D., Martins, M.M., Gherardini, P.F., Prestwood, T.R., Chabon, J., Bendall, S.C., et al. (2017). Systemic Immunity is required for effective cancer immunotherapy. *Cell* 168, 487–502.e15. <https://doi.org/10.1016/j.cell.2016.12.022>.
 20. Zhang, C., Bockman, A., and DuPage, M. (2024). Breaking up the CD8⁺ T cell: Treg pas de deux. *Cancer Cell* 42, 941–942. <https://doi.org/10.1016/j.ccell.2024.05.016>.
 21. Faustman, D.L., Torrey, H., Butterworth, J., Mera, T., Okubo, Y., Wang, L., Baum, D., Warden, S., Plager, S., Huang, D., et al. (2017). Targeting the TNFR2 oncogene on tumor cells and treigs of the tumor microenvironment via dominant TNFR2 antagonism. *J. Clin. Oncol.* 35, 106–106. https://doi.org/10.1200/JCO.2017.35.7_suppl.106.
 22. Dunn, M.R., Jimenez, R.M., and Chaput, J.C. (2017). Analysis of aptamer discovery and technology. *Nat. Rev. Chem.* 1. <https://doi.org/10.1038/s41570-017-0076>.
 23. Zhu, G., and Chen, X. (2018). Aptamer-based targeted therapy. *Adv. Drug Deliv. Rev.* 134, 65–78. <https://doi.org/10.1016/j.addr.2018.08.005>.
 24. Sun, Y., Mo, L., Hu, X., Yu, D., Xie, S., Li, J., Zhao, Z., Fang, X., Ye, M., Qiu, L., et al. (2022). Bispecific aptamer-based recognition-then-conjugation strategy for PD1/PDL1 axis blockade and enhanced immunotherapy. *ACS Nano* 16, 21129–21138. <https://doi.org/10.1021/acsnano.2c09093>.
 25. Menon, A.P., Villanueva, H., Meraviglia-Crivelli, D., van Santen, H.M.V., Hellmeier, J., Zheleva, A., Nonateli, F., Peters, T., Wachsmann, T.L.A., Hernandez-Rueda, M., et al. (2024). CD3 aptamers promote expansion and persistence of tumor-reactive T cells for adoptive T cell therapy in cancer. *Mol. Ther. Nucleic Acids* 35, 102198. <https://doi.org/10.1016/j.omtn.2024.102198>.
 26. Wang, Z., Zhang, Y., Wu, L., Chen, J., Xie, S., He, J., Zhang, Q., Chen, H., Chen, F., Liu, Y., et al. (2023). An aptamer-functionalized DNA circuit to establish an artificial interaction between T cells and cancer cells. *Angew. Chem. Int. Ed. Engl.* 62, e202307656. <https://doi.org/10.1002/anie.202307656>.
 27. Zhang, D., Zheng, Y., Lin, Z., Liu, X., Li, J., Yang, H., and Tan, W. (2020). Equipping natural killer cells with specific targeting and checkpoint blocking aptamers for enhanced adoptive immunotherapy in solid tumors. *Angew. Chem. Int. Ed. Engl.* 59, 12022–12028. <https://doi.org/10.1002/anie.202002145>.
 28. Ci, T., Li, H., Chen, G., Wang, Z., Wang, J., Abdou, P., Tu, Y., Dotti, G., and Gu, Z. (2020). Cryo-shocked cancer cells for targeted drug delivery and vaccination. *Sci. Adv.* 6, eabc3013. <https://doi.org/10.1126/sciadv.abc3013>.
 29. Liu, F., Xin, M., Feng, H., Zhang, W., Liao, Z., Sheng, T., Wen, P., Wu, Q., Liang, T., Shi, J., et al. (2024). Cryo-shocked tumor cells deliver CRISPR-Cas9 for lung cancer regression by synthetic lethality. *Sci. Adv.* 10, eadk8264. <https://doi.org/10.1126/sciadv.adk8264>.
 30. Zhao, N., Pei, S.N., Parekh, P., Salazar, E., and Zu, Y. (2014). Blocking interaction of viral gp120 and CD4-expressing T cells by single-stranded DNA aptamers. *Int. J. Biochem. Cell Biol.* 57, 10–18. <https://doi.org/10.1016/j.biocel.2014.03.008>.
 31. Lifson, J.D., Reyes, G.R., McGrath, M.S., Stein, B.S., and Engleman, E.G. (1986). AIDS retrovirus induced cytopathology: giant cell formation and involvement of CD4 antigen. *Science* 232, 1123–1127. <https://doi.org/10.1126/science.3010463>.
 32. Zhang, Y., Dong, Q., Xiao, J., Fang, X., Huang, W., Li, Q., Chen, Z., Liu, H., and Tan, W. (2025). In-vivo polyvalent aptamer@protein-based nanocarrier with synergistic charge effect for high drug loading, high nuclease resistance, and high receptor accessibility. *CCS Chem.* 7, 1411–1423. <https://doi.org/10.31635/ccschem.024.202404442>.
 33. Parkins, K.M., Melo, K.P., Chen, Y.X., Ronald, J.A., and Foster, P.J. (2021). Visualizing tumour self-homing with magnetic particle imaging. *Nanoscale* 13, 6016–6023. <https://doi.org/10.1039/D0NR07983A>.
 34. Kim, M.Y., Oskarsson, T., Acharyya, S., Nguyen, D.X., Zhang, X.H.F., Norton, L., and Massagué, J. (2009). Tumor self-seeding by circulating cancer cells. *Cell* 139, 1315–1326. <https://doi.org/10.1016/j.cell.2009.11.025>.
 35. Johnson, L.R., Lee, D.Y., Eacret, J.S., Ye, D., June, C.H., and Minn, A.J. (2021). The immunostimulatory RNA RN7SL1 enables CAR-T cells to

- enhance autonomous and endogenous immune function. *Cell* 184, 4981–4995.e14. <https://doi.org/10.1016/j.cell.2021.08.004>.
36. Comberlato, A., Paloja, K., and Bastings, M.M.C. (2019). Nucleic acids presenting polymer nanomaterials as vaccine adjuvants. *J. Mater. Chem. B* 7, 6321–6346. <https://doi.org/10.1039/C9TB01222B>.
37. Rushdi, M.N., Pan, V., Li, K., Choi, H.-K., Travaglino, S., Hong, J., Griffiths, F., Agnihotri, P., Mariuzza, R.A., Ke, Y., et al. (2022). Cooperative binding of T cell receptor and CD4 to peptide-MHC enhances antigen sensitivity. *Nat. Commun.* 13, 7055. <https://doi.org/10.1038/s41467-022-34587-w>.
38. Kocyla, A.M., Czogalla, A., Wessels, I., Rink, L., and Krężel, A. (2024). A combined biochemical and cellular approach reveals Zn²⁺-dependent hetero- and homodimeric CD4 and Lck assemblies in T cells. *Structure* 32, 292–303.e7. <https://doi.org/10.1016/j.str.2023.11.013>.
39. Moldovan, M.-C., Sabbagh, L., Breton, G., Sékaly, R.-P., and Krummel, M.F. (2006). Triggering of T cell activation via CD4 dimers. *J. Immunol.* 176, 5438–5445. <https://doi.org/10.4049/jimmunol.176.9.5438>.
40. Hunder, N.N., Wallen, H., Cao, J., Hendricks, D.W., Reilly, J.Z., Rodmyre, R., Jungbluth, A., Gnjjatic, S., Thompson, J.A., and Yee, C. (2008). Treatment of metastatic melanoma with autologous CD4⁺ T cells against NY-ESO-1. *N. Engl. J. Med.* 358, 2698–2703. <https://doi.org/10.1056/NEJMoa0800251>.
41. Lu, Y.-C., Parker, L.L., Lu, T., Zheng, Z., Toomey, M.A., White, D.E., Yao, X., Li, Y.F., Robbins, P.F., Feldman, S.A., et al. (2017). Treatment of patients with metastatic cancer using a major histocompatibility complex class II-restricted T-cell receptor targeting the cancer germline antigen MAGE-A3. *J. Clin. Oncol.* 35, 3322–3329. <https://doi.org/10.1200/JCO.2017.74.5463>.
42. Chen, D., Varanasi, S.K., Hara, T., Traina, K., Sun, M., McDonald, B., Farsakoglu, Y., Clanton, J., Xu, S., Garcia-Rivera, L., et al. (2023). CTLA-4 blockade induces a microglia-Th1 cell partnership that stimulates microglia phagocytosis and anti-tumor function in glioblastoma. *Immunity* 56, 2086–2104.e8. <https://doi.org/10.1016/j.immuni.2023.07.015>.
43. Zhang, S.Y., Guo, L., Zhang, Z.W., Liu, X.Y., Chen, W.J., Wei, Y., Wang, X.J., and Wu, Q. (2024). Type-I protein arginine methyltransferase inhibition primes anti-programmed cell death protein 1 immunotherapy in triple-negative breast cancer. *Cancer* 130, 1415–1423. <https://doi.org/10.1002/cncr.35142>.
44. Niu, W.B., and Spradling, A.C. (2022). Mouse oocytes develop in cysts with the help of nurse cells. *Cell* 185, 2576–2590.e12. <https://doi.org/10.1016/j.cell.2022.05.001>.
45. Katzenelenbogen, Y., Sheban, F., Yalin, A., Yofe, I., Svetlichnyy, D., Jaitin, D.A., Bornstein, C., Moshe, A., Keren-Shaul, H., Cohen, M., et al. (2020). Coupled scRNA-seq and intracellular protein activity reveal an immunosuppressive role of TREM2 in cancer. *Cell* 182, 872–885.e19. <https://doi.org/10.1016/j.cell.2020.06.032>.